MIL-R-55566C(CR)

11 August 1986
SUPERSEDING
MIL-R-55566B(EL)
9 July 1969

MILITARY SPECIFICATION

REEL, CABLE RC-435( )/U

This specification is approved for use within the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of tubular steel reel designated as Reel, Cable RC-435( )/U. (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116 Preservation, Methods of.
MIL-M-10578 Metal Conditioner and Rust Remover, Phosphoric Acid Base.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U.S. Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5017 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8130

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MIL-M-13231 Marking of Electronic Items.
MIL-F-14072 Finishes for Ground Signal Equipment.
MIL-E-55585 Electronic Equipment and Parts.

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.
MIL-STD-170 Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-202 Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 Standard General Requirements for Electronic Equipment.
MIL-STD-2073 DOD Material, Procedures for Development and Application of Packaging Requirements.

2.1.2 Other Government Documents, Drawings, and Publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Electronics Command

SC-DL-69296 Reel, Cable RC-435( )/U.

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The reel which this specification covers is Reel, Cable RC-435( )/U which is used for storage and transportation of various types of wire, cable or cable assemblies.

3.2 Construction. The reel shall be constructed in accordance with the requirements of this specification and Drawing and Data List SC-DL-69296.

3.3 Welding. All welds shall be of ample size with thorough penetration, good fusion and stronger than the material which they join.

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3.3.1 Cleaning Prior to Welding. Surface to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease and other foreign material.

3.3.2 Process. Preheating shall be employed where distortion is likely to result from welding. Welds shall have a thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved.

3.3.3 Cleaning After Welding. Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to finishing steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed. Sand blasted items shall be finished or a protective coating shall be applied within four hours of cleaning.

3.4 Finish, Protective. The reel shall be given protective finish in accordance with equipment drawings. Finish shall be smooth and free from blowholes. (See 4.5) Finish shall be applied after satisfactory completion of tests specified in 4.7, 4.8, and 4.9.

3.5 Identification Marking. All reels shall bear identification markings in accordance with the applicable detailed drawing and MIL-M-13231. (See 3.2)

3.6 Hub Size. When tested as specified in 4.6, the axle shall pass through the hub without binding.

3.7 Alignment. When tested as specified in 4.7, the rim shall be no more than one-fourth of an inch from the perpendicular to the axle.

3.8 Drop. The reel shall show no evidence of permanent degradation (other than deformation) and no loosening of parts as a result of the drop test specified in 4.8. The permanent deformation shall not exceed by more than one-half inch the measurements made in 4.7.

3.9 Vibration. When tested as specified in 4.9, the reels shall show no evidence of permanent degradation (including no permanent deformation or loosening of parts).

3.10 Moisture Resistance. Upon completion of test specified in 4.10, there shall be no evidence of corrosion or any other form of deterioration.

3.11 Workmanship. The reels shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein: (See 4.5).

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Construction	3.2
Welding	3.3
Cleaning prior to welding	3.3.1
Process	3.3.2
Cleaning after welding	3.3.3
Finish	3.4
Marking	3.5

3.12 First Article Units. When specified in the contract, the contractor shall subject samples to first article inspection in accordance with paragraph 4.3.

3.13 Mechanical Safety. The system design shall include the safety provisions called for in paragraph entitled "Mechanical Hazards of Requirement 1, MIL-STD-454. Exposed edges shall be rounded sufficiently to minimize laceration/puncture hazards, the goal being a minimum radius of 0.04 inch (1 mm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement.

4.1.1 Responsibility for Compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of Inspection. The inspections specified herein are classified as follows:

a. Quality conformance inspection.

(1) Quality Conformance inspection of equipment before packaging. (See 4.4)

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(2) Packaging inspection. (See 4.11)

b. First article inspection. (See 4.3)

4.3 First Article Inspection. This inspection shall consist of the inspection specified in Table I and shall include, as referenced on inspection data sheets, inspection specified in subsidiary documents and the supplementary tests. Order and quantity of equipments to be subjected to First Article Testing shall be as specified in Table I.

4.3.1 Standard of Workmanship. Standard of workmanship, 3.11, shall be established during the Visual Inspection of 4.5.

4.4 Quality Conformance Inspection of Equipment before Packaging. The contractor, to demonstrate compliance with specified requirements shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government reserves the right to review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. The contractor's inspection records for all inspections performed shall accurately reflect the observed attributes (actual measurements shall be recorded if otherwise required by the contract or this specification) for each characteristic identified within the required inspection/test parameters.

4.4.1 Group A Inspection.

4.4.1.1 Inspection Lot. An inspection lot shall consist of all reels of the same type, produced under essentially the same conditions and offered for inspection at one time.

4.4.1.2 Group A Tests. Group A tests shall consist of the tests specified in Table II, and shall be made on the same set of sample units in the order shown.

4.4.1.2.1 Sampling Plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection Level II. The acceptable quality level (AQL) shall be as specified in Table II. Major and minor defects shall be defined in MIL-STD-105.

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Table I - FIRST ARTICLE TEST SCHEDULE

Title	Req Para	Test Para	Sequence	
			#1	#2
Workmanship	3.11	4.5	1	1
Hub size	3.6	4.6	3	
Alignment	3.7	4.7	2	2
Drop	3.8	4.8		3
Vibration	3.9	4.9		4
Moisture Resistance	3.10	4.10	4	

Table II - GROUP A INSPECTION

Inspection	Req Para	Test Method Para	AQL	
			Major	Minor
Workmanship	3.11	4.5	1.0%	4.0%
Hub size	3.6	4.6	see note	

NOTE: All defects considered major, AQL is 1%.

4.4.2 Group B Inspection. This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of MIL-STD-105. The AQL shall be 1.0 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection.

Table III - GROUP B INSPECTION

Inspection	Req Para	Test Method Para
Alignment	3.7	4.7

4.4.3 Group C Inspection. This inspection shall consist of the tests specified in Table IV, and shall be performed on samples that have been subjected to and met Group A and Group B inspection. Sample units shall be selected at random without regard to their quality except that the samples selected at the start of production shall be selected from the first units produced. The tests specified in Table IV shall be performed at the beginning of production and according to the frequency specified, (months or units) whichever comes first.

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Table IV - GROUP C INSPECTION

Inspection	Req Para	Insp Para	Frequency Months	Frequency Units
Drop	3.8	4.8	1	1,000
Vibration	3.9	4.9	3	2,500
Moisture Resistance	3.10	4.10	6	5,000

4.4.3.1 Noncompliance. The contractor shall immediately report, in writing, each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections or the inspections which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.5)

4.4.3.2 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection, may be accepted on contract, provided that they are resubjected to and pass Group A and Group B inspection after repair of all visible damage.

4.5 Workmanship. The reels shall be given a thorough visual and mechanical inspection to ascertain that the materials, construction, identification marking, finish, and workmanship are in accordance with applicable requirements (See 3.11)

4.6 Hub Size. After finish a 1-1/32 + 0 -1/64 by 1-132 + 0 - 1/64 inch axle shall be inserted through the reel hub at two angles so that all grooves are tested to insure compliance with 3.6.

4.7 Alignment. Prior to applying finish the requirements of 3.7 shall be met when a 1-1/32 + 0 -1/64 by 1-1/32 + 0 -1/64 inch axle is inserted into the hub of the reel. Measurements shall be made at four points 90o apart on each rim.

4.8 Drop. The reel prior to finishing shall be loaded with a weight and quantity of cable comparable to its normal use as follows: 55 to 65 pounds including weight of reel and loaded to 80 to 90% of capacity.

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4.8.1 The reel shall be raised by a hoist equipped with a "quick release" device, to a height of 3-1/2 feet above a platform built of two inch fir backed by a concrete floor. By tripping the "quick release" device the reel shall be dropped squarely on the periphery of both rims.

4.8.2 The reel shall then be raised to a height of 3-1/2 feet and held so that when dropped, it will land squarely on one side. Then repeat on the other side.

4.8.3 The reel shall then be raised to a height of 3-1/2 feet and canted so that one rim is more than one inch higher then the other, so that when dropped it will land at an angle to the periphery of one rim but in an area opposite the opening on each rim. Then repeat on the other side.

4.8.4 At the conclusion of the tests, the reels shall be examined for compliance with 3.8.

4.9 Vibration. Prior to applying finish, the reels shall be subjected to the tests of 4.9.1 thru 4.9.4. During vibration the unit shall be clamped in its normal operating position on a vibration table that can be controlled to within 10 percent of the specified frequencies and amplitudes.

4.9.1 Direction of Vibration. The reels shall be vibrated successively, over the specified ranges of applied frequencies, in three mutually perpendicular directions parallel to the edges of the reels.

4.9.2 Rate of Change of Vibration Frequency. The frequency of the applied vibration shall be varied uniformly at a rate of approximately one Hz. The entire frequency range, from 10 to 33 cps and return to 10 cps, shall be traversed in approximately one minute.

4.9.3 Amplitude and Frequency of Applied Vibration. The amplitude (one-half of the total excursion) of the vibration table and the frequency range shall be in accordance with the following:

Frequency of Vibration (Hz)	Amplitude of Vibration (in)	
	Horizontal	Vertical
10 to 33	0.020	0.030

4.9.4 Duration of Test. The reels shall be vibrated for at least 90 minutes, 30 minutes in each of the three directions specified in 4.9.1.

4.9.5 At the conclusion of the test, the reels shall be examined for compliance with 3.9.

4.10 Moisture Resistance. The reel shall be subjected to 15 continuous 48-hour cycles. Temperature, relative humidity, and period of time for each

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portion of the cycle shall conform to MIL-STD-170. The equipment shall meet the requirements of 3.10.

4.11 Packaging Inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.11.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.11.2 Quality Conformance Inspection of Packaging.

4.11.2.1 Materials Inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.11.2.2 Preservation Inspection. Inspection of preservation and interior markings shall be in accordance with Group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.11.2.3 Packing Inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table V "PACKING INSPECTION PROVISIONS". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

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Table V. PACKING INSPECTION PROVISIONS.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING.

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073, coded as follows:

5.1.1 Level A. 10-1-00-00-NS-X-ED

5.1.2 Level B. 10-1-00-00-NS-X-ED

5.2 Packing and Marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES.

6.1 Intended Use. The reel covered by this specification is intended for use in storing and transporting wire, cable or cable assemblies.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Type required.

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c. Level A or B preservation and packing (see 5).

d. Special marking required, if any. (see 5.2)

e. Place of final inspection.

f. When first article rough handling tests are not required.

g. When first article packaging inspection reports require acquisition activity approval prior to production unit packaging.

6.3 Level B Preservation. When Level B preservation is specified, this level of protection will only be used under known favorable conditions during transporation, storage, and handling.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, Reel, Cable RC-435W/U. As soon as possible after the award of the contract, the contractor should apply to the government office specified in the contract for such information. (See 1.1)

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specification referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Group C Inspection. Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.4.3.1)

6.7 Subject Term (Keyword) Listing.

Alignment
Cable
Distribution Statement A
Finish Protective
Hub Size
Mechanical Safety
Moisture Resistance
Nomenclature
Reel, Cable RC-435( )/U
Tubular Steel Reel
Vibration
Welding

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6.8 Changes From Previous Issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing Activity:

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Project No. 8130-A016

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street City State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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